Towards Feed-forward 3D Reconstruction

Yu-Chun Lin

Contents

What is 3D Reconstruction?

The process of creating a digital three-dimensional model of an object or scene from data. It turns (2D) observations into a usable 3D model.

Videos
(ordered/unordered) Images
Depth (Kinect, LiDAR)?
IMU?
…

Point cloud: A set of 3D points
Meshes: Points connected into triangles
Voxel: A 3D pixel
NeRF, 3D Gaussians, …

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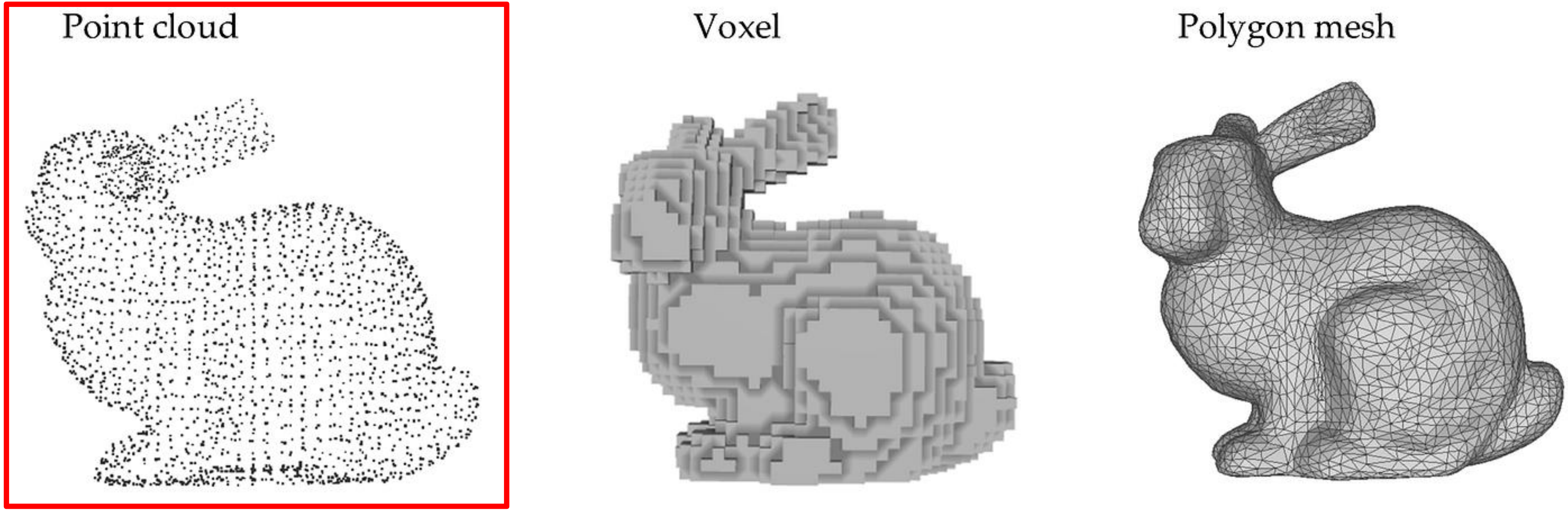
The process of creating a digital three-dimensional model of an object or scene from data. It turns (2D) observations into a usable 3D model.



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Overview

3D Reconstruction

- Structure-from-Motion (SfM) Offline
  - COLMAP
- Simultaneous Localization and Mapping (SLAM) Online
  - ORB-SLAM 2/3
  - DSO
  - LSD-SLAM
- Others

- Input: Images
- Output: Point Cloud

3D Reconstruction Methods

Deep Learning

Optimization-based

Incremental:
COLMAP

Global:
GLOMAP

Optimization-based

With Global Alignment:
DUSt3R, MASt3R

Scene Coordinate Reconstruction:
ACE0

Others:
FlowMap, VGGSfM

Feed-forward

VGGT-based:
SAIL-Recon

DUSt3R-based:
Cut3R, Fast3R

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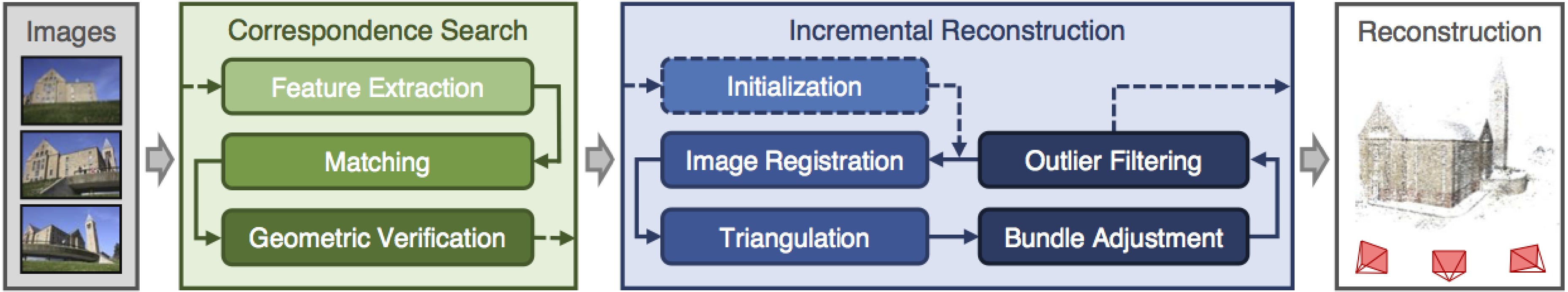
COLMAP

- A system for SfM + Multi-View Stereo (MVS)
- Input: (Un)ordered images
- Output: Point Cloud & Estimated camera poses



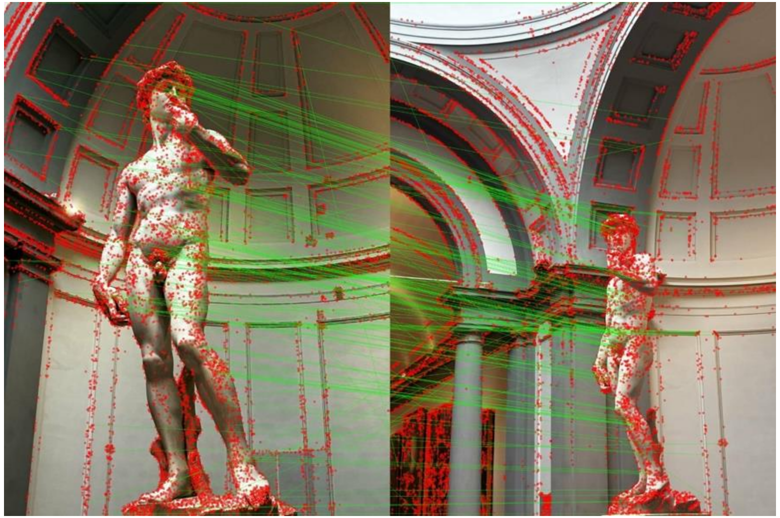
https://colmap.github.io/tutorial.html

COLMAP: Feature Matching

- SIFT (Scale-Invariant Feature Transform)
  - 128-dimension descriptors of keypoints
- Compare each descriptor of A with all descriptors of B
- Use L2 distance to find the nearest neighbor
- Lowe's ratio test

$$\frac{||\mathrm{d_A} - \mathrm{d_1}||}{||\mathrm{d_A} - \mathrm{d_2}||} < 0.7$$

$\mathrm{d_A}$: Descriptor
$\mathrm{d_1}$: The nearest neighbor
$\mathrm{d_2}$: The second nearest neighbor



COLMAP: Incremental Reconstruction

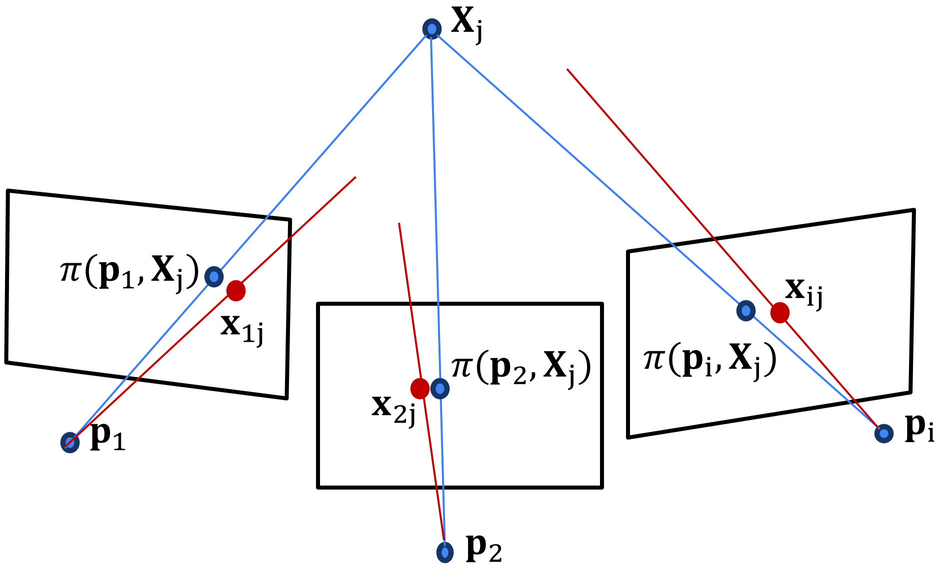
- Initialization: Camera poses & 3D points
- Select next image to be added (i.e. overlap)
- Solve for camera pose of the added image
- Triangulate new 3D points
- Bundle adjustment

https://colmap.github.io/tutorial.html

COLMAP: Bundle Adjustment

- Least squares approach to jointly estimate camera poses and 3D points
- Minimize "reprojection error"

$$\min \sum_{(\mathrm{i,j})} ||\mathbf{x}_{\mathrm{ij}} - \pi(\mathbf{p}_{\mathrm{i}}, \mathbf{X}_{\mathrm{j}})||^2$$



$\mathbf{x}_{\mathrm{ij}}$: The 2D observation (pixel) of 3D point $\mathbf{X}_{\mathrm{j}}$ in camera i

$\pi(\mathbf{p}_{\mathrm{i}}, \mathbf{X}_{\mathrm{j}})$: The projection function that maps the 3D point $\mathbf{X}_{\mathrm{j}}$ into the 2D pixel location predicted by parameters $\mathbf{p}_{\mathrm{i}}$

COLMAP: Bundle Adjustment

- Least squares approach to jointly estimate camera poses and 3D points
- Minimize "reprojection error"

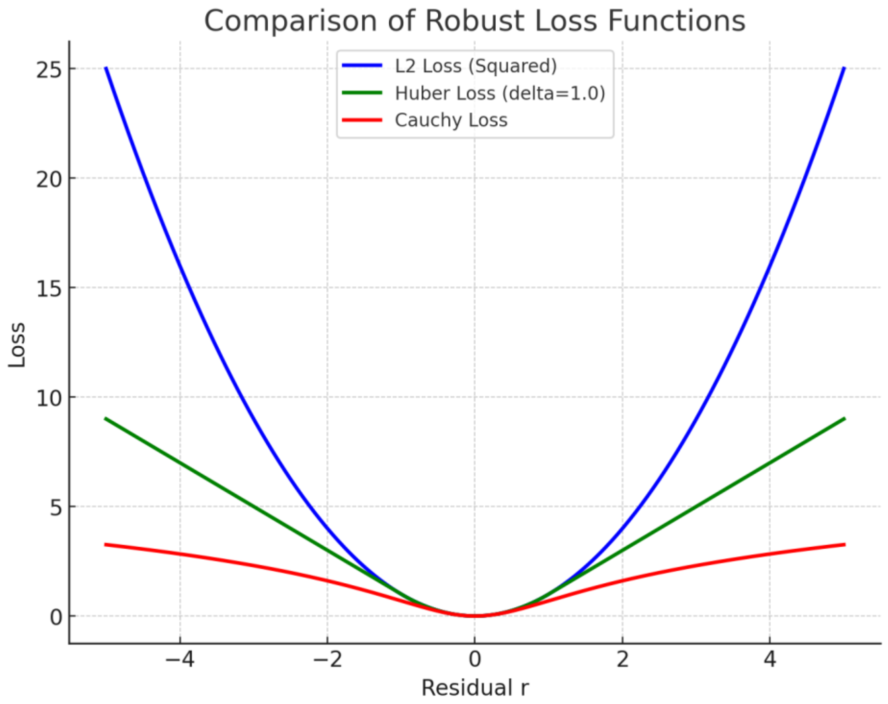
$$\min \sum_{(i,j)} \rho(||\mathbf{x}_{ij} - \pi(\mathbf{p}_i, \mathbf{X}_j)||^2)$$

Residual $r_{ij} = \mathbf{x}_{ij} - \pi(\mathbf{p}_i, \mathbf{X}_j)$

$\mathbf{x}_{ij}$: The 2D observation (pixel) of 3D point $\mathbf{X}_j$ in camera i

$\pi(\mathbf{p}_i, \mathbf{X}_j)$: The projection function that maps the 3D point $\mathbf{X}_j$ into the 2D pixel location predicted by parameters $\mathbf{p}_i$

$\rho(\cdot)$: Robust kernel



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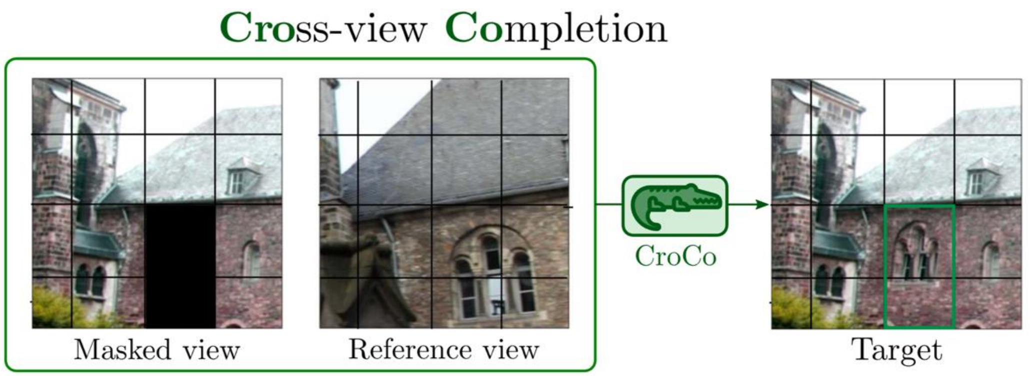
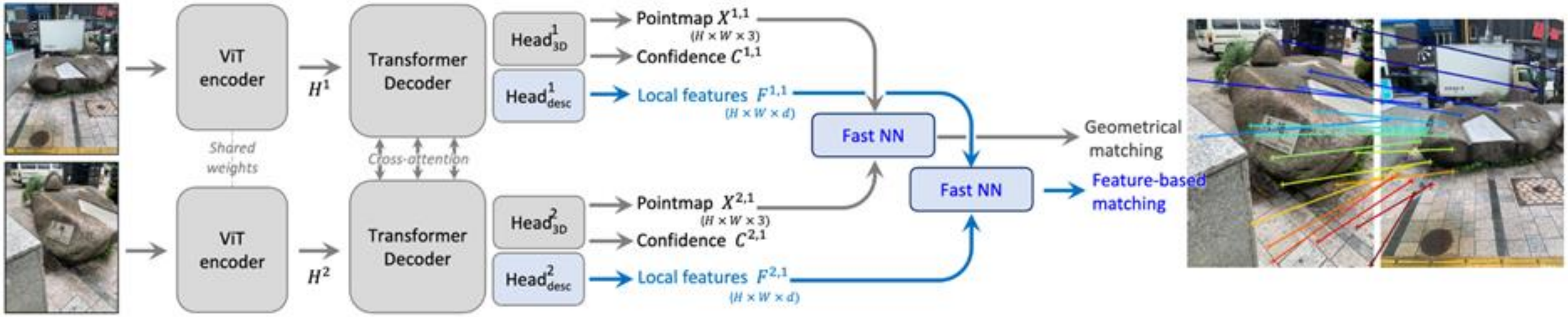
DUSt3R (MASt3R)

- CroCo pre-trained encoder
- Input: “Two” images
- Output: Pointmap, Confidence, Local features
- Image Matching in 3D



https://arxiv.org/pdf/2210.10716



https://arxiv.org/pdf/2406.09756

DUSt3R (MASt3R)

Given a set of images $\{\mathrm{I}^1, \mathrm{I}^2, \dots, \mathrm{I}^N\}$,

=> Construct a connectivity graph $\mathcal{G}(\mathcal{V}, \mathcal{E})$

$$\chi^* = \underset{\chi, P, \sigma}{\arg\min} \sum_{e \in \mathcal{E}} \sum_{v \in e} \sum_{i=1}^{HW} C_i^{v,e} \left\| \chi_i^v - \sigma_e P_e X_i^{v,e} \right\|$$

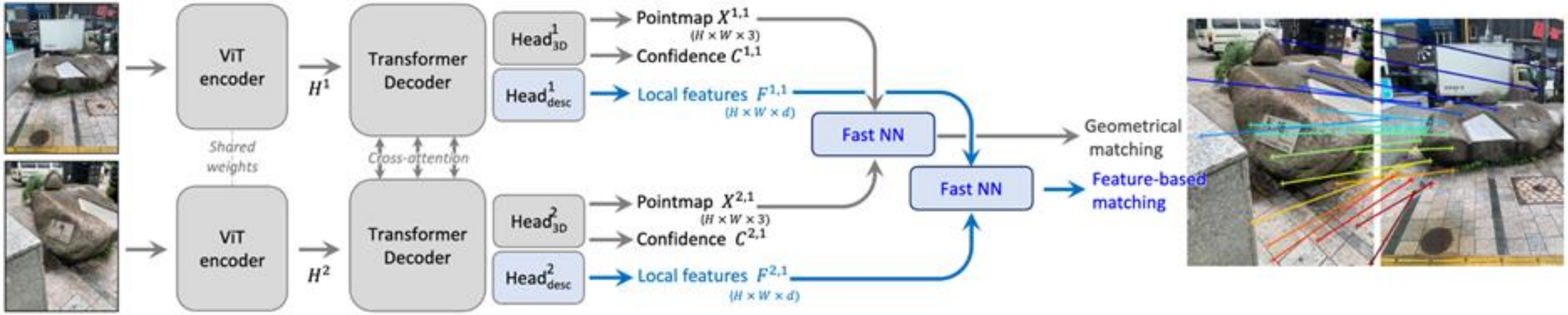
$e = (n, m) \in \mathcal{E}$: Image pair

P_e: Rigid Transform

$X_i^{v,e}$: Pointmap (local coordinate)

χ_i^v: Pointmap (world coordinate)

σ_e: Scaling



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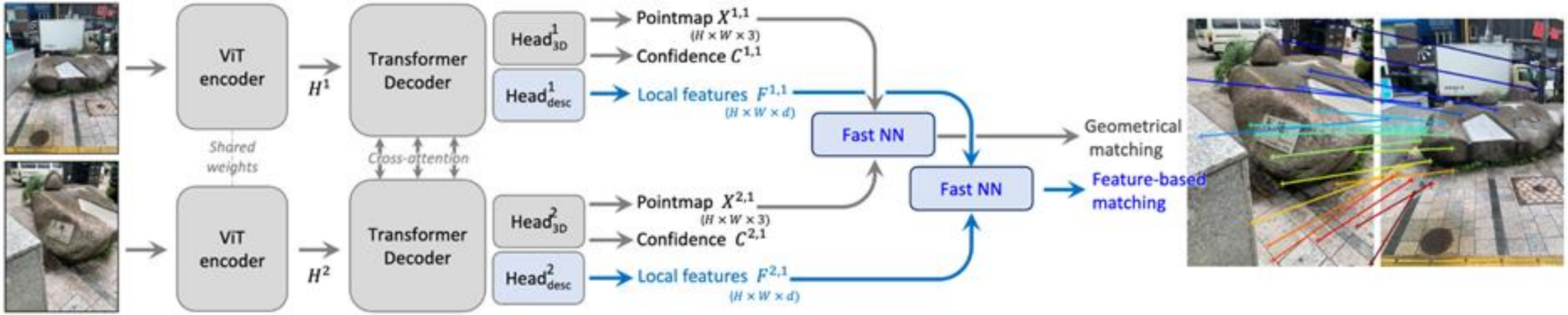

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Feed-forward

VGGT-based:
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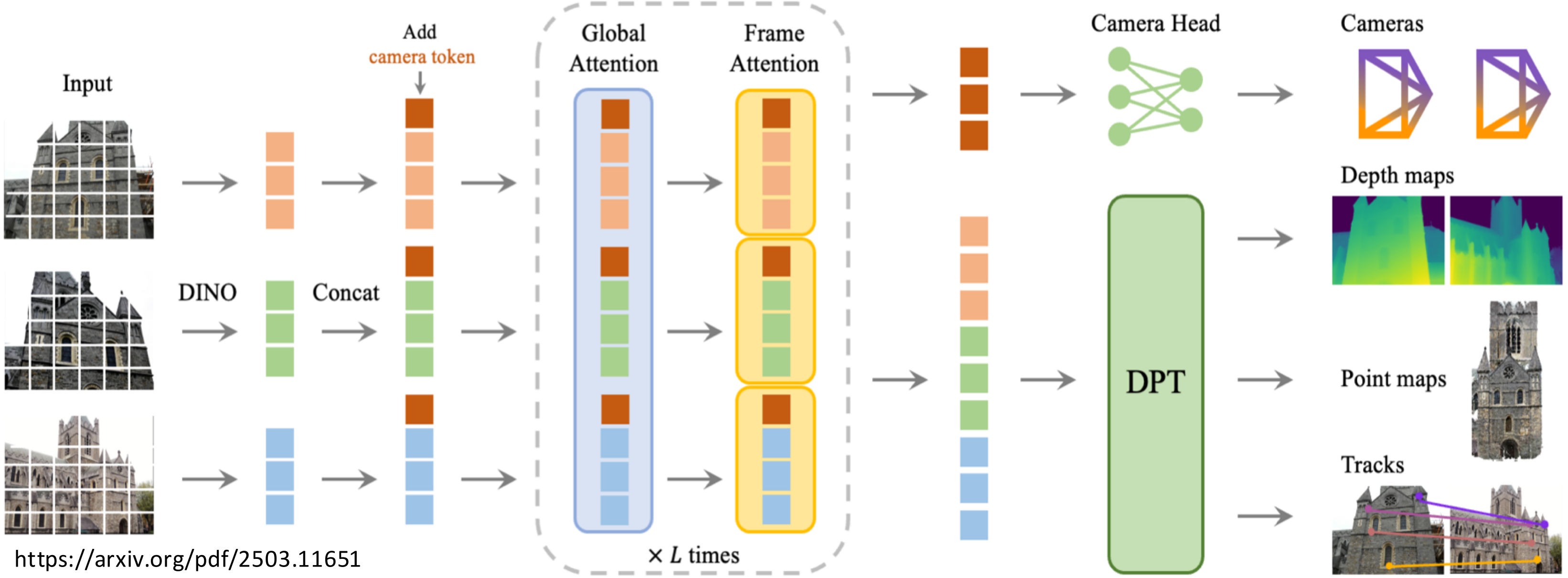
VGGT

- Feed-forward method
- Input: (Un)ordered Images
- Output: Camera parameters, point maps, depth maps, and 3D point tracks

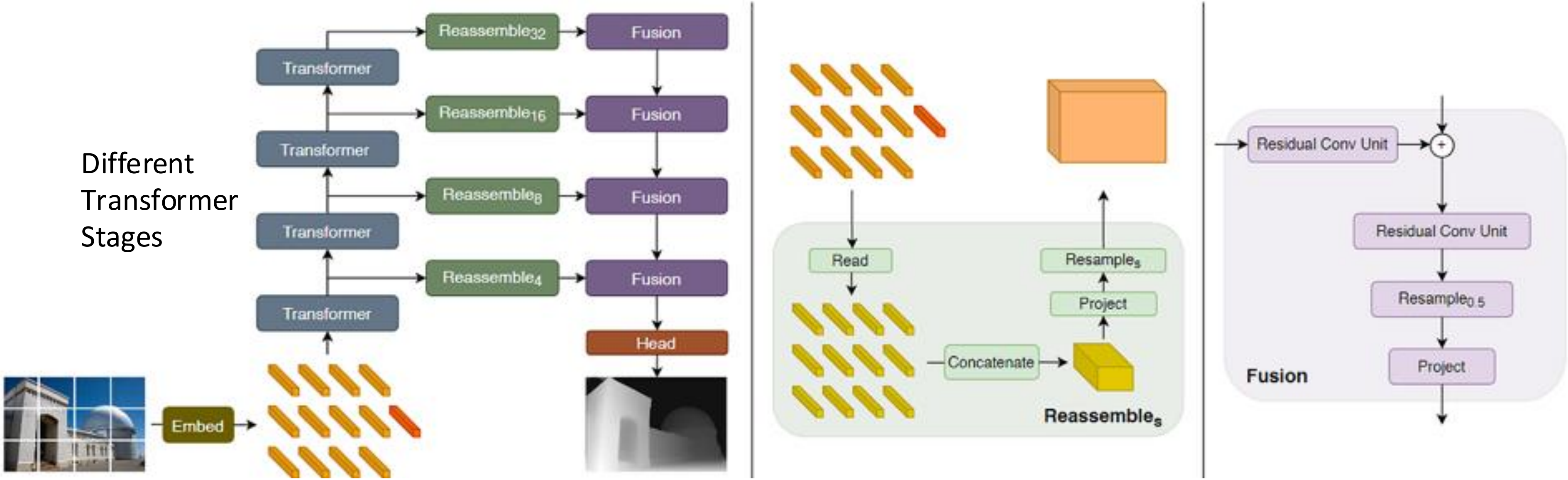


https://arxiv.org/pdf/2503.11651

VGGT

- Dense prediction transformer (DPT)
- Embed: Flattening non-overlapping patches + linear projection (same as ViT)

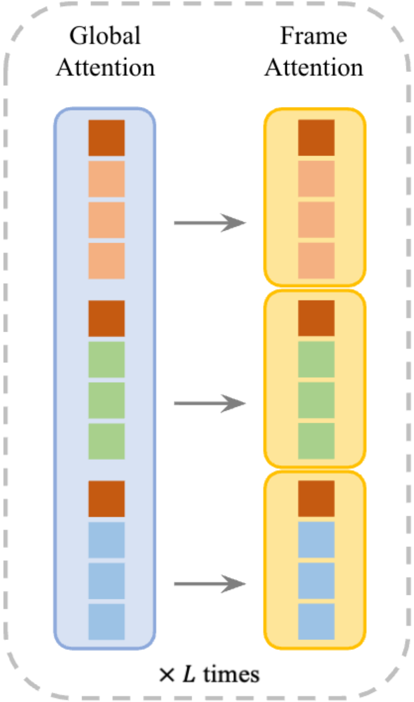
Different Transformer Stages



https://arxiv.org/pdf/2103.13413

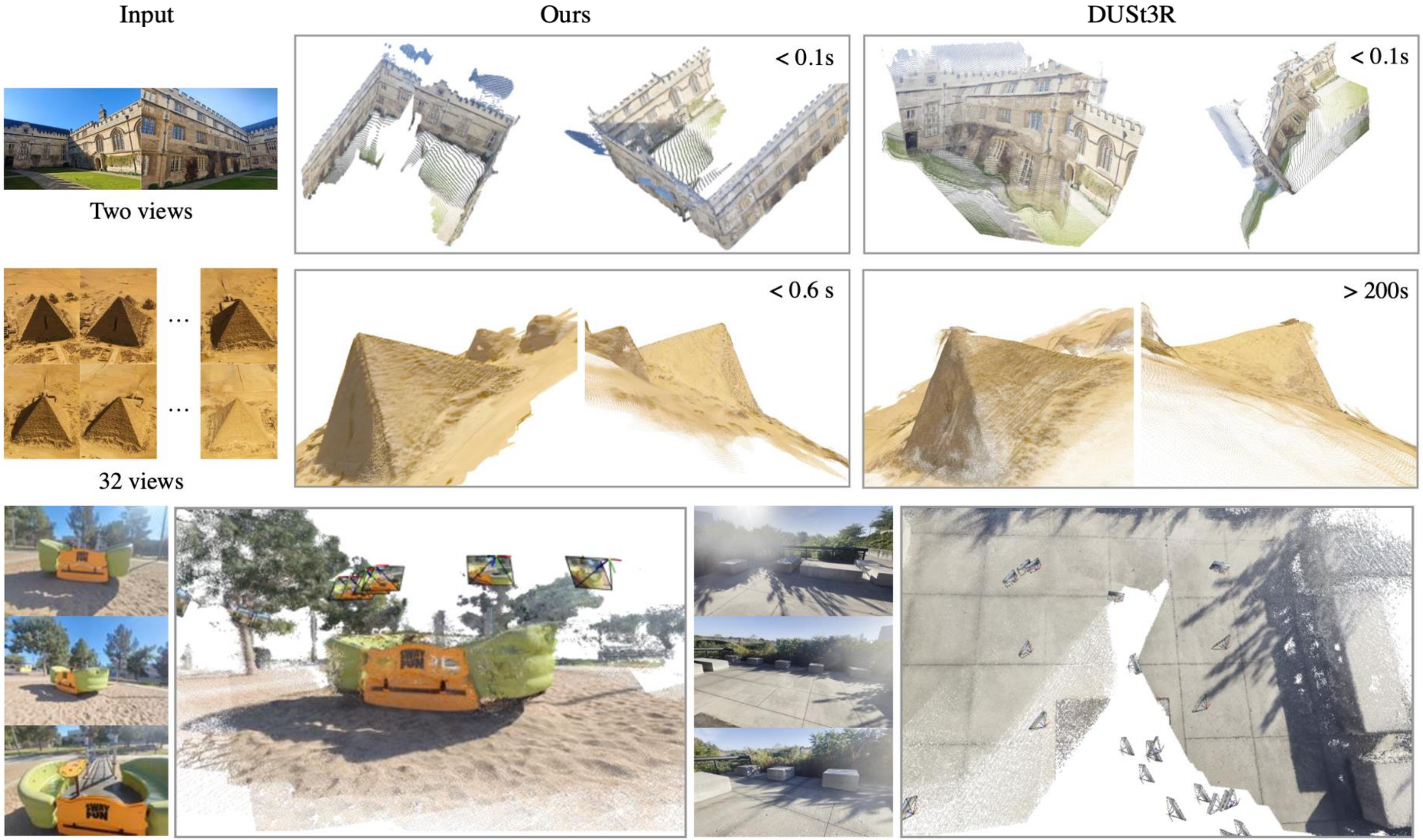
VGGT: Alternating-Attention

- Global attention
    - Ensures scene-level coherence
- Frame-wise attention
    - Eliminates **frame index embedding**
        - For permutation equivariance (input -> output)
        - For flexible input length



https://docs.google.com/presentation/d/1JVuPnuZx6RgAy-U5Ezobg73XpBi7FrOh/edit?slide=id.p9#slide=id.p9

VGGT



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Feed-forward 3D Reconstruction

Strengths

- Speed (cf. Bundle Adjustment, Global Alignment)
- Learned priors from data
- Reduced error accumulation

Weaknesses

- Generalization
- Accuracy

Current Problems & Perspectives

- Scalability (OOM)

  - Fast3R

- Large-scale 3D Reconstruction

  - SAIL-Recon

- Online 3D Reconstruction

  - Cut3R

Fast3R

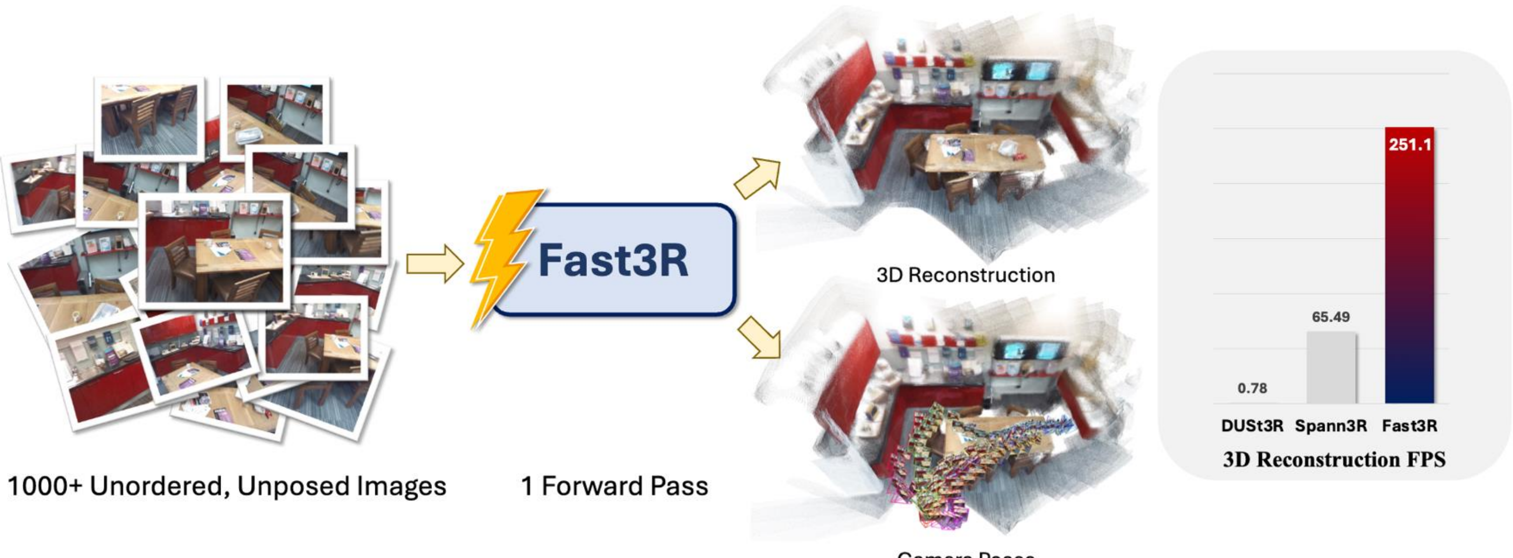
Fast3R: Towards 3D Reconstruction of 1000+ Images in One Forward Pass

Jianing Yang♣♦ Alexander Sax♣ Kevin J. Liang♣ Mikael Henaff♣ Hao Tang♣
Ang Cao♣♦ Joyce Chai♦ Franziska Meier♣ Matt Feiszli♣

♣ Meta FAIR ♦ University of Michigan

- Scalability (OOM)

  - Fast3R



https://arxiv.org/pdf/2501.13928

Fast3R

- Observation

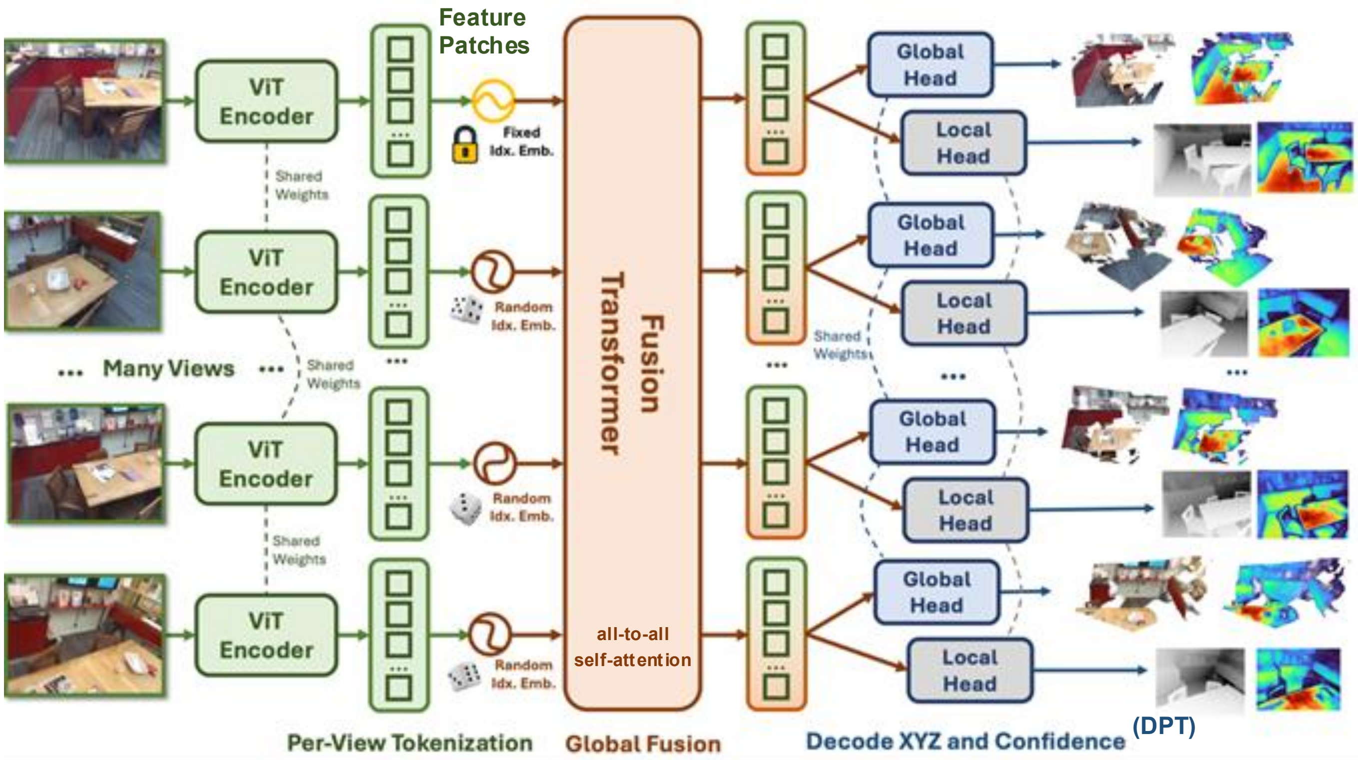
1. DUSt3R leads to OOM with only 48 views on an A100 GPU

2. DUSt3R computes $O(N^2)$ pairs of pointmaps and performs a global alignment optimization procedure

3. DUSt3R suffers from the same pair-wise bottleneck as traditional SfM

Fast3R

Inputs: Unordered, unposed, RGB images $\mathbf{I} \rightarrow (\mathbf{X}_L, \Sigma_L, \mathbf{X}_G, \Sigma_G)$



Global Pointmap $\mathbf{X}_G$:
In the coordinate frame of the 1st camera

Local Pointmap $\mathbf{X}_L$:
In the coordinate frame of the viewing camera

Align $\mathbf{X}_L$ to $\mathbf{X}_G$ using ICP
=> Aligned local pointmaps
=> 3D reconstruction

https://arxiv.org/pdf/2501.13928

Fast3R: Technical Highlights

- Training Objective: $\mathcal{L}_{\text{total}} = \mathcal{L}_{\mathbf{X}_\text{G}} + \mathcal{L}_{\mathbf{X}_\text{L}}$ Confidence-weighted versions of the <u>normalized 3D pointwise regression loss</u>

$$\ell_{\text{regr}}(\hat{\mathbf{X}}, \mathbf{X}) = \left\| \frac{1}{\hat{z}}\hat{\mathbf{X}} - \frac{1}{z}\mathbf{X} \right\|_2, \qquad z = \frac{1}{|\mathbf{X}|}\sum_{x \in \mathbf{X}} \|x\|_2$$

Dataset: CO3D, ScanNet++, ARKitScenes, Habitat, BlendedMVS, MegaDepth (less than DUSt3R)

- Fast3R can handle more views at inference than were used to train a model

Naive way: Spherical Harmonic

- Trainging: SH({1, 2, …, N})
- Testing: SH({1, 2, …, N_{test}})

Adoption: Position Interpolation

- Draw N ⊂ {1, 2, …, N’} indexes uniformly at random (N’ ≫ N)

Masking!

Fast3R: Experiments

Inference Efficiency

# Views	Fast3R Time (s)	Fast3R Peak GPU Mem (GiB)	DUSt3R Time (s)	DUSt3R Peak GPU Mem (GiB)
2	0.065	3.84	0.092	3.52
8	0.122	6.33	8.386	24.59
32	0.509	13.25	129.0	67.61
48	0.84	20.8	OOM	OOM
320	15.938	41.90	OOM	OOM
800	89.569	55.97	OOM	OOM
1000	137.62	63.01	OOM	OOM
1500	308.85	78.59	OOM	OOM

3D Reconstruction

Method	FPS	7 scenes [47] Acc↓	7 scenes [47] Comp↓	NRGBD [3] Acc↓	NRGBD [3] Comp↓
F-Recon [65]	<0.1	7.62	2.31	20.59	6.31
DUSt3R† [61]	0.78	**1.23**	0.91	**2.51**	1.03
Spann3R [57]	65.4	1.48	**0.85**	3.15	1.10
Fast3R (Ours)	**251.1**	1.58	0.93	3.40	**1.01**

Skip = 20 Skip = 40

Method	Views	DTU Acc↓ Med.	DTU Comp↓ Med.
DUSt3R [61]	all/5	**1.159**	0.914
DUSt3R† [61]	all/5	1.297	1.002
Spann3R [57]	all/5	2.268	1.295
Fast3R (Ours)	all/5	1.706	**0.857**

https://arxiv.org/pdf/2501.13928

Fast3R: Experiments

Pose Estimation

Methods	CO3Dv2 [39]					RealEstate10K [75]	FPS
	RRA@15↑	RRA@5↑	RTA@15↑	RTA@5↑	mAA(30)↑	mAA(30)↑	
Colmap+SG [9, 41]	36.1	24.4	27.3	17.2	25.3	45.2	0.056
PixSfM [28]	33.7	26.1	32.9	17.6	30.1	49.4	-
RelPose [71]	57.1	-	-	-	-	-	0.02
PosReg [59]	53.2	-	49.1	-	45.0	-	0.015
PoseDiff [59]	80.5	59.5	79.8	61.7	66.5	48.0	0.015
RelPose++ [27]	(85.5)	-	-	-	-	-	0.02
DUSt3R [60]	96.2	-	86.8	-	76.7	67.7	0.78
MASt3R [23]	94.6	93.2	**91.9**	**86.2**	81.8	**76.4**	0.23
Fast3R-no-outdoor (Ours)	**99.7**	**97.4**	87.1	76.1	**82.5**	-	**251.1**
Fast3R (Ours)	96.2	90.2	81.6	68.2	75.0	72.7	**251.1**

https://arxiv.org/pdf/2501.13928

SAIL-Recon

- Large-scale 3D Reconstruction

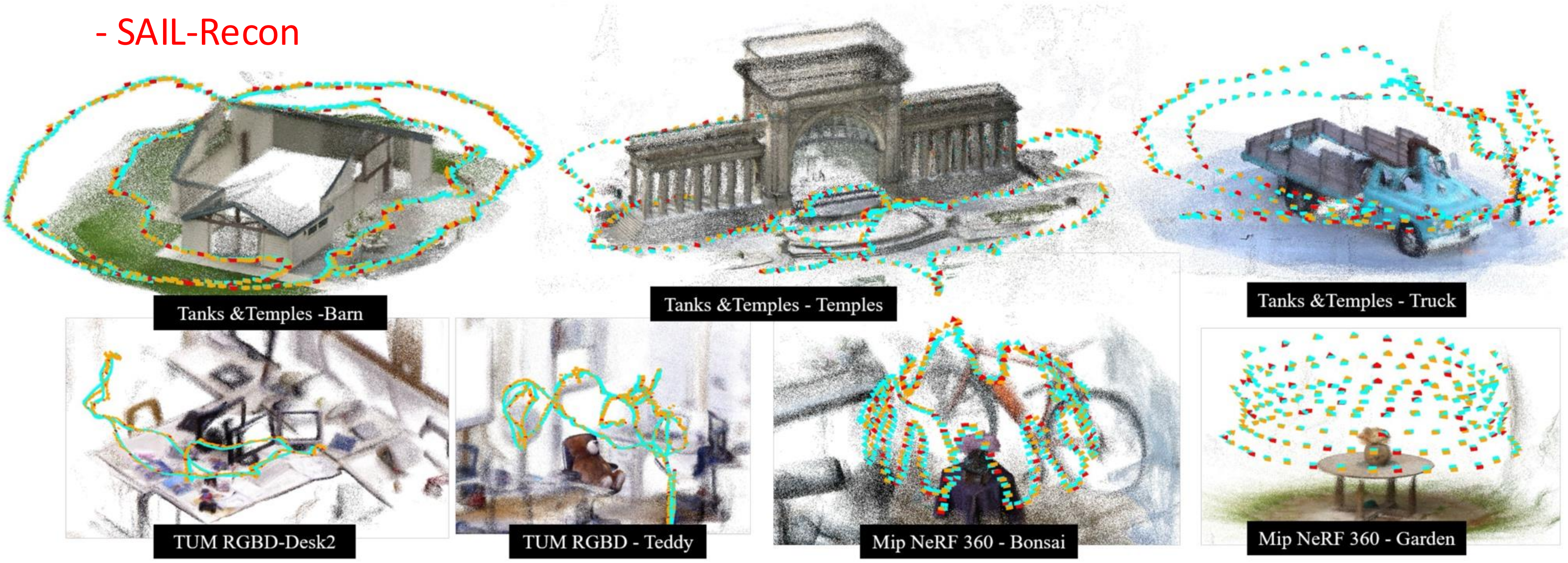
  - SAIL-Recon

Based on VGGT

SAIL-Recon: Large SfM by Augmenting $\boxed{\text{Scene Regression}}$ with Localization

Junyuan Deng[1,2*] Heng Li[1*] Tao Xie[2,3] Weiqiang Ren[2] Qian Zhang[2]
Ping Tan[1†] Xiaoyang Guo[2†]

[1]The Hong Kong University of Science and Technology [2]Horizon Robotics [3]Zhejiang University



https://arxiv.org/pdf/2508.17972

SAIL-Recon

- Observation

1. VGGT struggle to handle a large number of input images

2. Current large-scale, incremental reconstruction methods suffer from pose drift and are highly dependent on subsequent global alignment (e.g. online, divide and conquer methods)

3. Existing SCR methods ignore visual localization, a common approach to scale up an SLAM system

SAIL-Recon

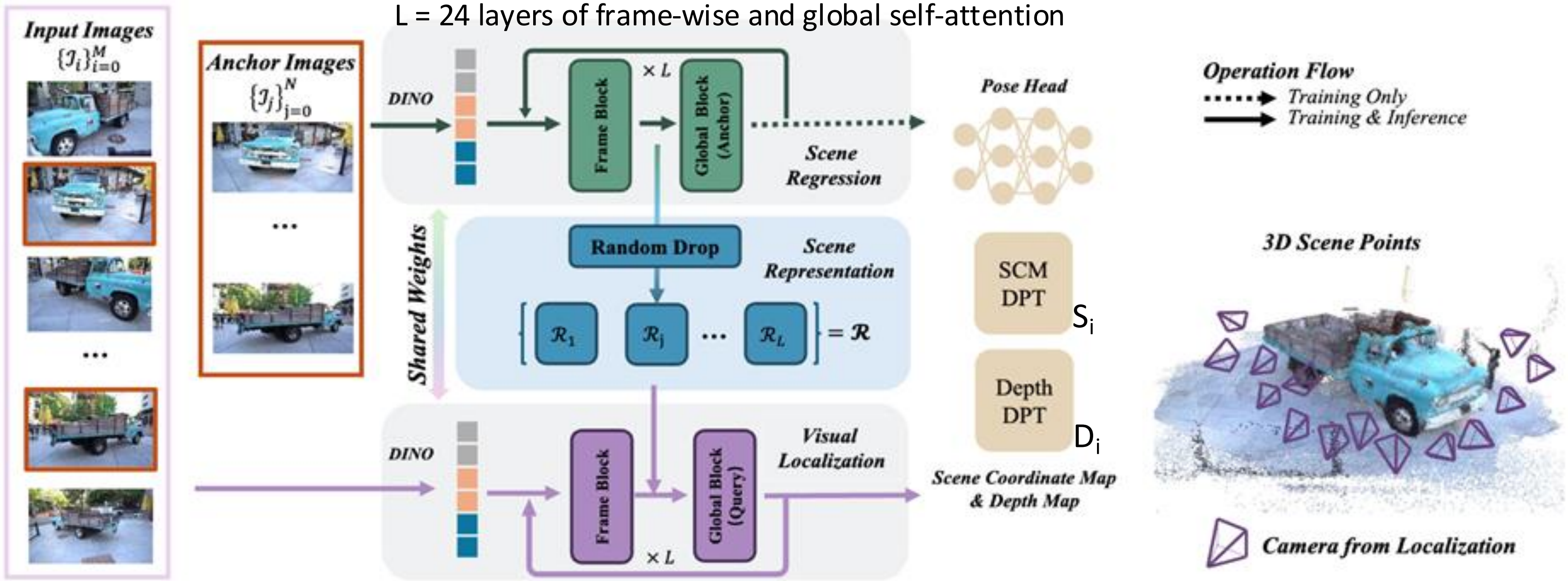
$$(\mathcal{R}, \{T_i, K_i, D_i, S_i\}_{i=1}^{M}) = \mathcal{T}_\theta(\{\mathcal{I}_i\}_{i=1}^{M})$$

$$\{T^q, K^q, D^q, S^q\} = \mathcal{T}_\theta(\mathcal{I}^q, \mathcal{R})$$

$\mathcal{T}_\theta$: Transformer-based network
T_i : Extrinsic matrices
K_i : Intrinsic matrices
$\mathcal{I}^q$: Query image

Inputs: Unordered, unposed, RGB images

L = 24 layers of frame-wise and global self-attention



SAIL-Recon: Technical Highlights

- Neural scene representation serves as an implicit neural map of the scene, without relying on explicit 3D points or meshes

- Most localization methods require per-scene training of the network
  => SAIL-Recon circumvents this by deriving a latent scene representation through scene regression

- Uniformly sample N ∈ [50, 100] anchor frames $\{\mathcal{I}_i\}_{i=1}^{N}$ from the full set $\{\mathcal{I}_i\}_{i=1}^{M}$

SAIL-Recon: Technical Highlights

The network $\mathcal{T}_\theta$ takes a set of 2D images as input and progressively enhances them to compute 3D structures

=> Extract intermediate feature tokens from each attention layer of $\mathcal{T}_\theta$ to form the scene representation R

(Captures the gradual evolution from 2D appearance features to 3D coordinate descriptors)

$$\mathcal{R} = [\{\Theta(t_j^{'\mathcal{I}})\}_{j=1}^{L}]$$

$\boldsymbol{\Theta}$: Downsampling operation

$t_j^{'\mathcal{I}}$: Intermidate feature tokens

Randomly select a ratio r ∈ [0.2, 1.0]

=> Sample a subset of tokens in each anchor frame to control the total size of $\Theta(t_j^{'\mathcal{I}})$

SAIL-Recon: Technical Highlights

Method	Co3Dv2↑ mAA@30	mAA@5	RRA@15	RTA@15
SAIL-Recon	**87.3**	**57.6**	**98.3**	**93.6**
300 per image → Fixed token	86.5	53.6	98.2	93.6
Avg. Pooling	86.5	53.5	98.2	93.5

Table 7. **Ablation on Training Strategy.** We investigate the different training strategies on pose estimation accuracy.

$$\mathcal{R} = [\{\Theta(t_j^{'\mathcal{I}})\}_{j=1}^{L}]$$

Θ : Downsampling operation

$t_j^{'\mathcal{I}}$: Intermidate feature tokens

Randomly select a ratio r ∈ [0.2, 1.0]

=> Sample a subset of tokens in each anchor frame to control the total size of $\Theta(t_j^{'\mathcal{I}})$

SAIL-Recon: Technical Highlights

- Training Objective: $\mathcal{L} = \mathcal{L}_{\text{camera}} + \mathcal{L}_{\text{depth}} + \mathcal{L}_{\text{scm}}$
  (Fine-tuning VGGT checkpoints)

$$\mathcal{L}_{\text{camera}} = \sum_{i=1}^{N} \|\hat{g}_i - g_i\|_1$$

$$\mathcal{L}_{\text{depth}} = \sum_{i=1}^{N} \|C_i^D \odot (\hat{D}_i - D_i)\| + \|(\nabla \hat{D}_i - \nabla D_i)\| - \alpha \log C_i^D$$

Gradient-based smooth

$$\mathcal{L}_{\text{scm}} = \sum_{i=1}^{N} \|C_i^S \odot (\hat{S}_i - S_i)\| + \|(\nabla \hat{S}_i - \nabla S_i)\| - \alpha \log C_i^S$$

With normalization w.r.t. a randomly selected image in anchor frames

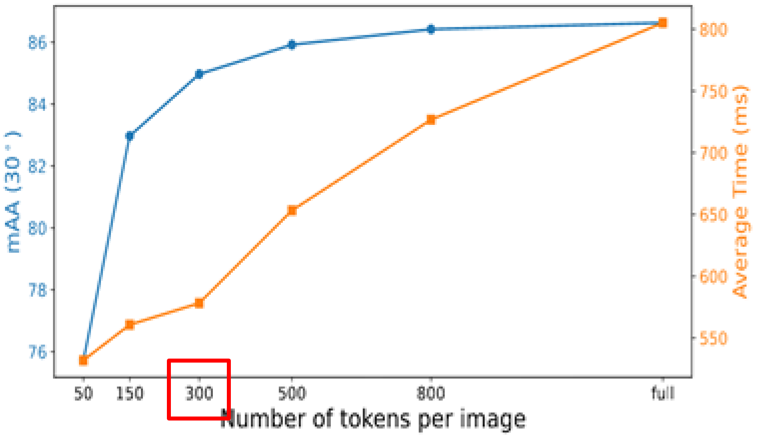
C_i^D: Uncertainty Map

$g_i = [\mathbf{q}, \mathbf{t}, \mathbf{f}]$:
Camera Parameters
q: Camera orientation
t: Translation vector
f: Field of View

SAIL-Recon: Experiments

Pose Accuracy - Tanks & Temples.

Method	Align.	RRA@5 ↑	RTA@5 ↑	ATE ↓	Reg. ↑	Time [s] ↓
COLMAP [60]	OPT	GT	GT	GT	GT	-
GLOMAP [48]	OPT	75.8	76.7	0.010	100.0	1977
ACE0 [9]	OPT	56.9	57.9	0.015	100.0	5499
DF-SfM [26]	OPT	69.6	69.3	0.014	76.2	-
FlowMap [63]	OPT	31.7	35.7	0.017	66.7	-
VGGSfM [84]	OPT	-	-	-	0.0	2134
MASt3R-SfM [19]	OPT	49.2	54.0	0.011	100.0	2723
DROID-SLAM [75]	OPT	31.3	40.3	0.021	100.0	240
SAIL-Recon-OPT	OPT	71.5	77.7	0.008	100.0	233
Cut3R† [86]	FFD	18.8	25.8	0.017	100.0	42
Spann3R† [82]	FFD	22.1	30.7	0.016	100.0	116
SLAM3R† [42]	FFD	20.3	24.7	0.015	100.0	70
Light3R-SfM [21]	FFD	52.0	52.8	0.011	100.0	63
VGGT-SLAM*△ [43]	FFD	57.3	67.9	0.008	100.0	238
SAIL-Recon	FFD	70.4	74.7	0.008	100.0	81



SAIL-Recon: Experiments

Pose Accuracy - Tanks & Temples.

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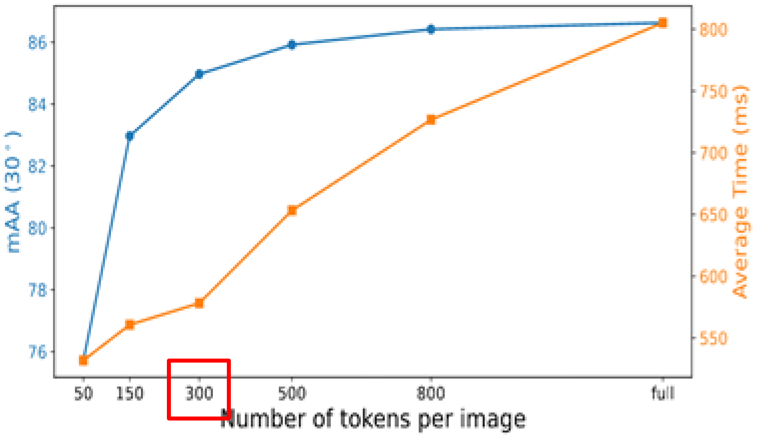
Light3R-SfM [21]	FFD	52.0	52.8	0.011	100.0	63
VGGT-SLAM*△ [43]	FFD	57.3	67.9	0.008	100.0	238
SAIL-Recon	FFD	70.4	74.7	0.008	100.0	81



The evaluation of camera poses is sometimes unreliable, as the pseudo ground-truth from COLMAP is only an estimation!

SAIL-Recon: Experiments

Pose accuracy via view synthesis on Mip-NeRF 360

PSNR

	Pseudo GT COLMAP	DROID-SLAM† [75]	BARF [37]	Nope-NeRF [5]	ACE0 [9]	Ours
Bicycle	21.5	10.9	11.9	12.2	18.7	20.50
Bonsai	27.6	10.9	12.5	14.8	25.8	26.76
Counter	25.5	12.9	11.9	11.6	24.5	25.51
Garden	26.3	16.7	13.3	13.8	25.0	24.92
Kitchen	27.4	13.9	13.3	14.4	26.1	27.43
Room	28.0	11.3	11.9	14.3	19.8	27.46
Stump	16.8	13.9	15.0	13.9	20.5	20.83
Average	24.7	12.9	12.8	13.5	22.9	24.77
Average Time	1h	2min	4h	≥24h	8h	5min

SAIL-Recon: Ablation

Method	Global Align.	Co3Dv2↑		
		RRA@15	RTA@15	mAA@30
Colmap [59]	OPT	31.6	27.3	25.3
Glomap [48]	OPT	45.9	40.3	37.3
PixSfM [38]	OPT	33.7	32.9	30.1
VGGSfM [84]	OPT	92.1	88.3	74.0
DUSt3R-GA [87]	OPT	96.2	86.8	76.7
MASt3R-SfM [19]	OPT	96.0	93.1	88.0
PoseDiff [83]	FFD	80.5	79.8	66.5
RelPose++ [36]	FFD	82.3	77.2	65.1
Spann3R [82]	FFD	89.5	83.2	70.3
MASt3R * [34]	FFD	94.5	80.9	68.7
Light3R-SfM [21]	FFD	94.7	85.8	72.8
VGGT [85]	FFD	98.4	94.8	88.2
SAIL-Recon $N = 10$	FFD	98.3	94.0	88.1
SAIL-Recon $N = 8$	FFD	98.2	93.6	87.3
SAIL-Recon $N = 5$	FFD	97.7	92.2	85.0
SAIL-Recon $N = 2$	FFD	96.4	89.7	78.5

Ablation on the number of anchor views for pose estimation performance on **Co3Dv2**

# of input images: 10

Demo

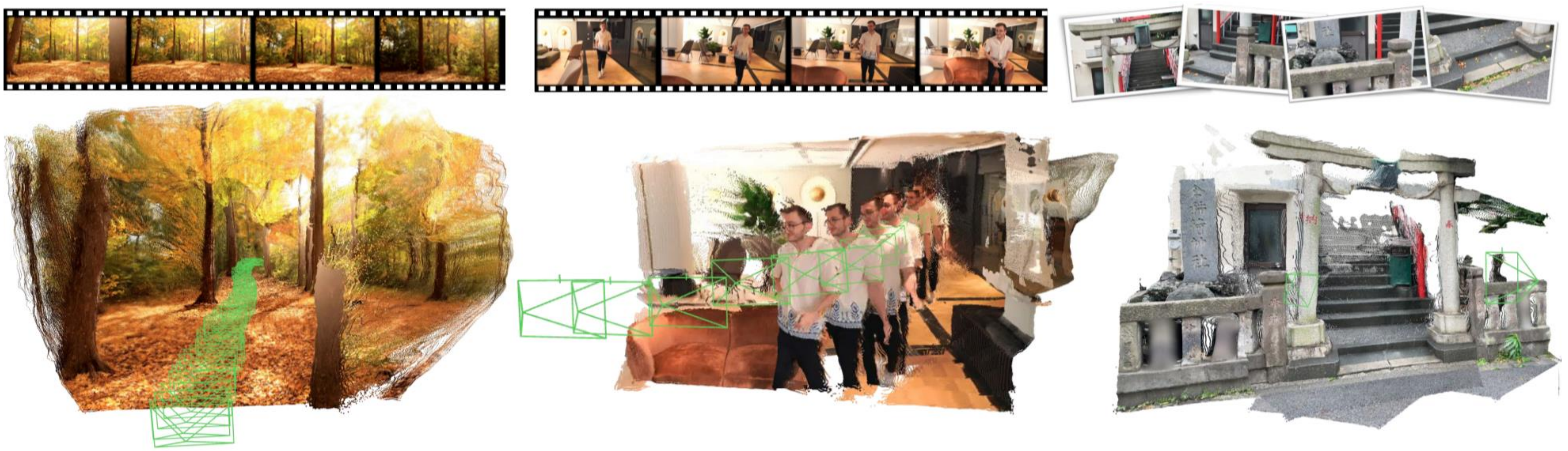
Cut3R

Continuous 3D Perception Model with Persistent State

Qianqian Wang[1,2*], Yifei Zhang[1*], Aleksander Holynski[1,2], Alexei A. Efros[1], Angjoo Kanazawa[1]
[1]University of California, Berkeley [2]Google DeepMind

- Online 3D Reconstruction

  - Cut3R



Static Scene Dynamic Scene Sparse Photo Collection

https://arxiv.org/pdf/2501.12387

Cut3R

- Observation

  1. Many 3D reconstruction pipelines operate as tabula rasa, starting from scratch for each new scene (e.g. SfM, SLAM, …)

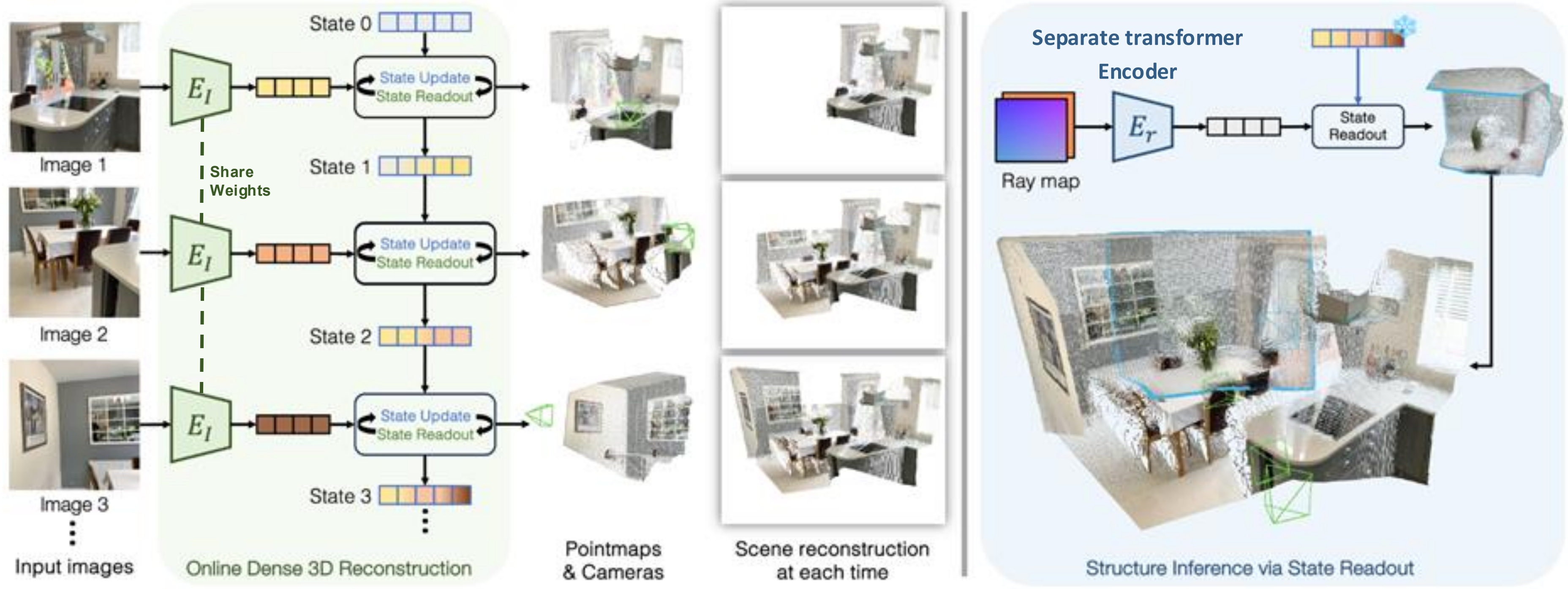
  => Poses challenges in handling scenarios with sparse observations

  2. Current continuous reconstruction methods have limitation such as being object-centric, or limited to static scenes

Cut3R

Ray map **R**:
A 6-channel image encoding the origin and direction of rays at each pixel

Inputs: Stream of images without any camera information



Poses

https://arxiv.org/pdf/2501.12387

Cut3R: Technical Highlights

- The bidirectional interaction (update and readout) is implemented using two interconnected transformer decoders

$$[\boldsymbol{z}'_t, \boldsymbol{F}'_t], \boldsymbol{s}_t = \text{Decoders}([\boldsymbol{z}, \boldsymbol{F}_t], \boldsymbol{s}_{t-1})$$

$$\left.\begin{aligned} \hat{\boldsymbol{X}}_t^{\text{self}}, \boldsymbol{C}_t^{\text{self}} &= \text{Head}_{\text{self}}(\boldsymbol{F}'_t) \\ \hat{\boldsymbol{X}}_t^{\text{world}}, \boldsymbol{C}_t^{\text{world}} &= \text{Head}_{\text{world}}(\boldsymbol{F}'_t, \boldsymbol{z}'_t) \end{aligned}\right\} \text{DPT}$$

$$\hat{\boldsymbol{P}}_t = \text{Head}_{\text{pose}}(\boldsymbol{z}'_t), \quad \text{MLP}$$

F_t : Image tokens
z : Learnable pose token
X_t : Pointmap
C_t : Confidence map
P_t : (6-DoF) Relative transformation

- Redundancy simplifies training (i.e.predicting $\hat{\boldsymbol{X}}_t^{\text{self}}$, $\hat{\boldsymbol{X}}_t^{\text{world}}$, and $\hat{\boldsymbol{P}}_t$)
- All pointmaps and poses are in metric scale (i.e. meters)

Cut3R: Technical Highlights

- Training Objective

$$\mathcal{L}_{\text{conf}} = \sum_{(\hat{\boldsymbol{x}},c)\in(\hat{\mathcal{X}},\mathcal{C})} \left(c \cdot \left\| \frac{\hat{\boldsymbol{x}}}{\hat{s}} - \frac{\boldsymbol{x}}{s} \right\|_2 - \alpha \log c \right)$$

Confidence-aware regression loss to the pointmaps. (When GT pointmaps are metric: set $\hat{s} := s$)

$$\mathcal{L}_{\text{pose}} = \sum_{t=1}^{N} \left(\|\hat{\boldsymbol{q}}_t - \boldsymbol{q}_t\|_2 + \left\| \frac{\hat{\boldsymbol{\tau}}_t}{\hat{s}} - \frac{\boldsymbol{\tau}_t}{s} \right\|_2 \right)$$

Pose loss. We parameterize the pose $\hat{\boldsymbol{P}}_t$ as quaternion $\hat{\boldsymbol{q}}_t$ and translation $\hat{\boldsymbol{\tau}}_t$, and minimize the L2 norm between the prediction and groundtruth:

$$\mathcal{L}_{rgb} = ||\hat{\boldsymbol{I}}_r - \boldsymbol{I}_r||_2^2$$

MSE loss. When the input is raymap, we also apply MSE loss to enforce the predicted pixel colors $\hat{\boldsymbol{I}}_r$ to match the groundtruth

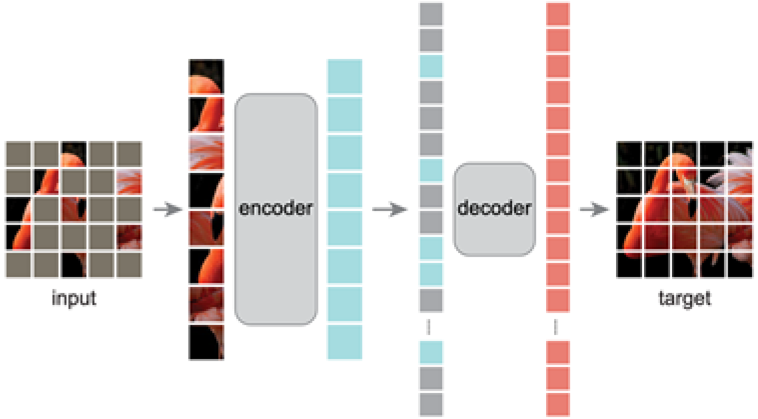
Cut3R: Analogy

Cut3R

- Completion occurs at the image level

- Using global context of the 3D scene captured in the state.

Masked Autoencoders



- Completion occurs at the patch level
- Using global context of the entire image

Cut3R: Experiments

Video Depth Estimation

Alignment	Method	Optim.	Onl.	Sintel Abs Rel ↓	Sintel δ<1.25 ↑	BONN Abs Rel ↓	BONN δ<1.25 ↑	KITTI Abs Rel ↓	KITTI δ <1.25 ↑	FPS
Per-sequence scale	DUSt3R-GA [107]	✓		0.656	45.2	0.155	83.3	<u>0.144</u>	<u>81.3</u>	0.76
	MASt3R-GA [51]	✓		0.641	43.9	0.252	70.1	0.183	74.5	0.31
	MonST3R-GA [125]	✓		**0.378**	**55.8**	**0.067**	**96.3**	0.168	74.4	0.35
	Spann3R [101]		✓	0.622	42.6	0.144	81.3	0.198	73.7	13.55
	Ours		✓	<u>0.421</u>	<u>47.9</u>	<u>0.078</u>	<u>93.7</u>	**0.118**	**88.1**	16.58
Metric scale	MASt3R-GA [51]	✓		**1.022**	14.3	0.272	70.6	0.467	15.2	0.31
	Ours		✓	1.029	**23.8**	**0.103**	**88.5**	**0.122**	**85.5**	16.58

1. Absolute relative error (Abs Rel)
2. $\delta < 1.25$: Percentage of predicted depths within a 1.25-factor of true depth

Cut3R: Experiments

Camera Pose Estimation

			Sintel			TUM-dynamics			ScanNet		
Method	**Optim.**	**Onl.**	ATE ↓	RPE trans ↓	RPE rot ↓	ATE ↓	RPE trans ↓	RPE rot ↓	ATE ↓	RPE trans ↓	RPE rot ↓
Particle-SfM [129]	✓		0.129	0.031	**0.535**	-	-	-	0.136	0.023	0.836
Robust-CVD [47]	✓		0.360	0.154	3.443	0.153	0.026	3.528	0.227	0.064	7.374
CasualSAM [128]	✓		0.141	**0.035**	0.615	0.071	**0.010**	1.712	0.158	0.034	1.618
DUSt3R-GA [107]	✓		0.417	0.250	5.796	0.083	0.017	3.567	0.081	0.028	0.784
MASt3R-GA [51]	✓		0.185	0.060	1.496	**0.038**	0.012	**0.448**	0.078	0.020	**0.475**
MonST3R-GA [125]	✓		**0.111**	0.044	0.869	0.098	0.019	0.935	**0.077**	**0.018**	0.529
DUSt3R [107]		✓	0.290	0.132	7.869	0.140	0.106	3.286	0.246	0.108	8.210
Spann3R [101]		✓	0.329	0.110	4.471	0.056	0.021	0.591	**0.096**	0.023	0.661
Ours		✓	**0.213**	**0.066**	**0.621**	**0.046**	**0.015**	**0.473**	0.099	**0.022**	**0.600**

3D Reconstruction

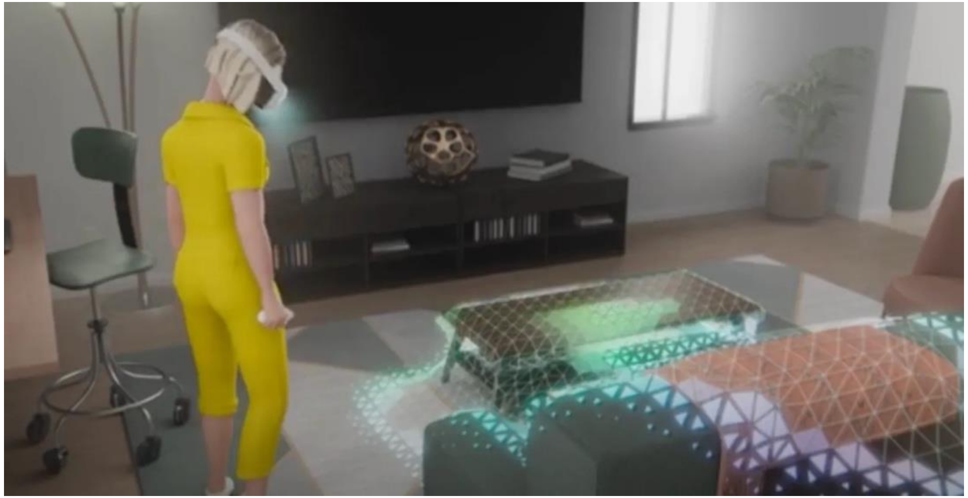
			7 scenes [83]						NRGBD [4]						
			Acc↓		Comp↓		NC↑		Acc↓		Comp↓		NC↑		
Method	**Optim.**	**Onl.**	Mean	Med.	Mean	Med.	Mean	Med.	Mean	Med.	Mean	Med.	Mean	Med.	**FPS**
DUSt3R-GA [107]	✓		0.146	0.077	0.181	0.067	**0.736**	**0.839**	0.144	**0.019**	0.154	**0.018**	**0.870**	**0.982**	0.68
MASt3R-GA [51]	✓		0.185	0.081	0.180	0.069	0.701	0.792	**0.085**	0.033	**0.063**	0.028	0.794	0.928	0.34
MonST3R-GA [51]	✓		0.248	0.185	0.266	0.167	0.672	0.759	0.272	0.114	0.287	0.110	0.758	0.843	0.39
Spann3R [101]		✓	0.298	0.226	0.205	0.112	0.650	0.730	0.416	0.323	0.417	0.285	0.684	0.789	12.97
Ours		✓	**0.126**	**0.047**	**0.154**	**0.031**	0.727	0.834	0.099	0.031	0.076	0.026	0.837	0.971	17.00

Demo

[Dynamic Scene 1]
[Dynamic Scene 2]
[Static Scene]

3D Reconstruction Application: AR/VR

- Apple ARKit – RoomPlan
- Meta Quest 3 spatial anchoring



https://www.meta.com/zhtw/help/quest/625635239532590/

3D Reconstruction Application: Medical Imaging

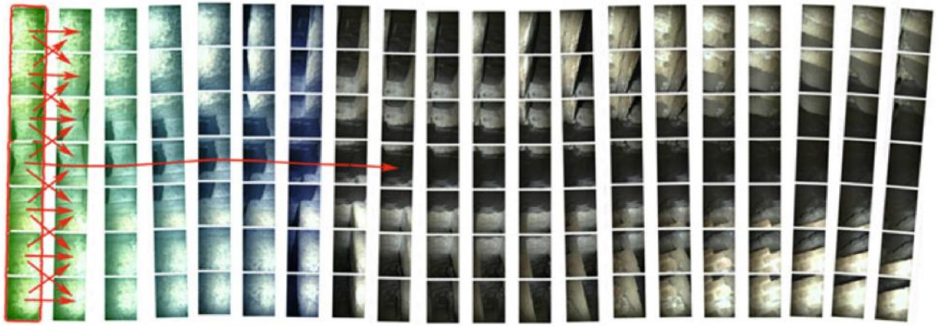
- Endoscopy surface reconstruction

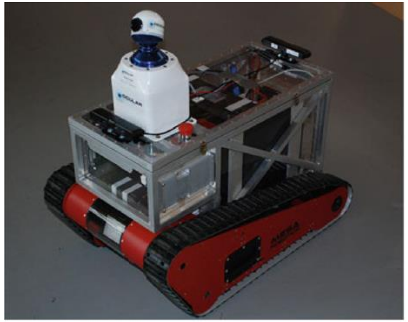
Endoscopy images from VR simulators: https://arxiv.org/pdf/2008.12949

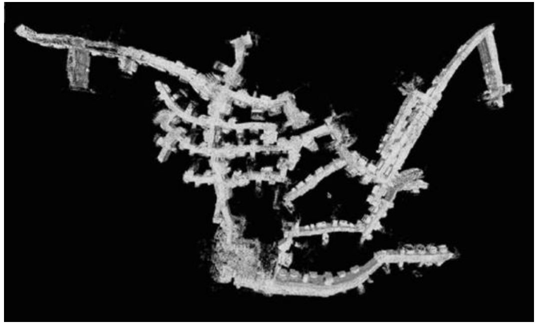
Surface reconstruction results from MPAC work

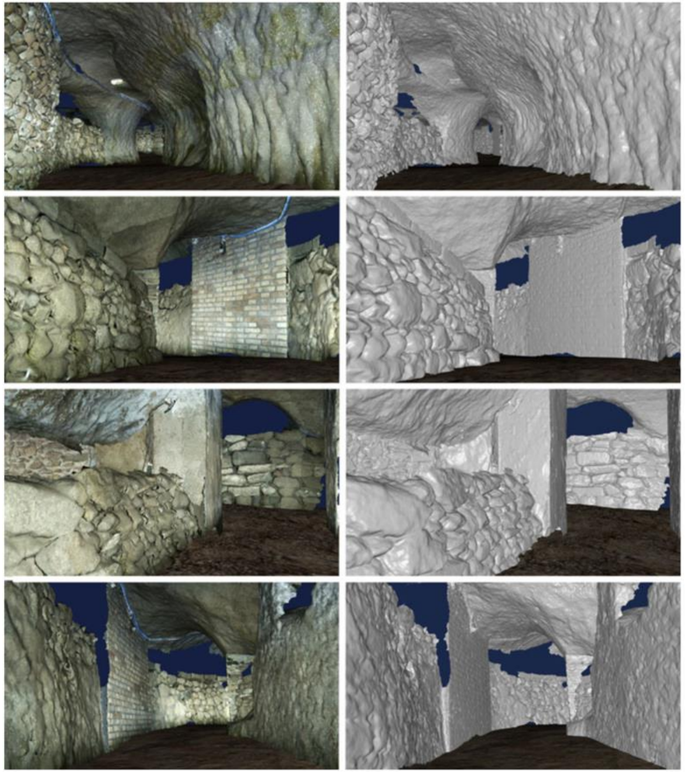
3D Reconstruction Application: Robotics

- Autonomous Mapping of the Priscilla Catacombs





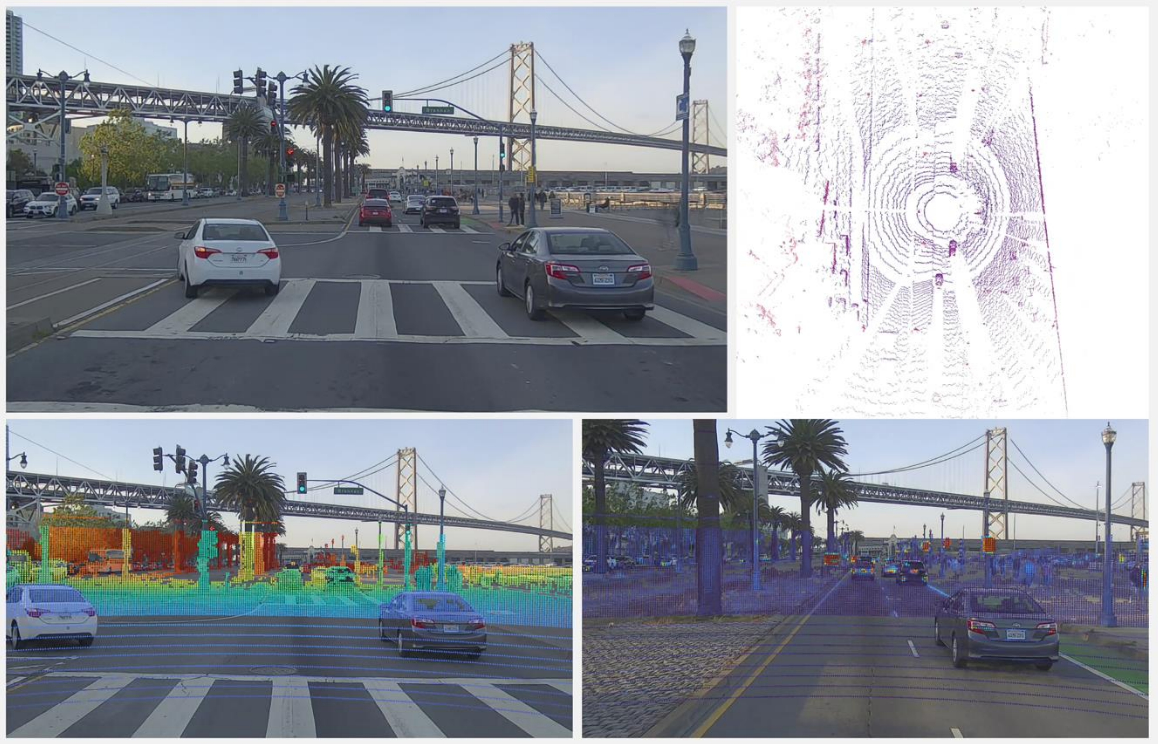




https://homes.esat.kuleuven.be/~konijn/publications/2017/978-3-319-49607-8_3.pdf

3D Reconstruction Application: Autonomous Driving

- SplatAD: Real-Time Lidar and Camera Rendering with 3D Gaussian Splatting for Autonomous Driving
  (2025 CVPR)



- Rendered images
- Bird's-eye views of the rendered point cloud
- Point clouds overlaid on images, with points colored based on range and intensity.

https://research.zenseact.com/publications/splatad/

References

[1] Qianqian Wang, Yifei Zhang, Aleksander Holynski, Alexei A. Efros, and Angjoo Kanazawa. Continuous 3d perception model with persistent state. (CVPR 2025 Oral)

[2] Jianing Yang, Alexander Sax, Kevin J. Liang, Mikael Henaff, Hao Tang, Ang Cao, Joyce Chai, Franziska Meier, and Matt Feiszli. Fast3R: Towards 3d reconstruction of 1000+ images in one forward pass. (CVPR 2025)

[3] Junyuan Deng, Heng Li, Tao Xie, Weiqiang Ren, Qian Zhang, Ping Tan, and Xiaoyang Guo. SAIL-Recon: Large SfM by augmenting scene regression with localization. (2025)